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RCPL rPET QR Code Information System

Product Requirements Document

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ORGANISATION	RCPL Pvt Ltd
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✓ Approved for Development

v1.1 · Dual QR System — FSSAI + PWM Declaration

Document Changelog

Version	Date	Author	Changes
`v1.0`	April 2026	Ashish Chandorkar	Initial release — FSSAI Product Information QR System (System 1)
`v1.1`	April 2026	Ashish Chandorkar	Added System 2: rPET Plastic Recycle Declaration QR System. Updated architecture, data model, API spec, UX requirements, acceptance criteria, and delivery plan. Fixed DB layer: SQLite → JSON file store.

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1. Executive Summary

Project: RCPL operates **two independent QR Code systems** within a single codebase, each serving a different regulatory requirement for their rPET product line. The system is fully independent from existing RLL/RCPM QR infrastructure.

System 1

FSSAI Product Information

QR code maps to `/`. Consumers enter a 1–2 character batch code from the bottle neck to retrieve manufacturer name, vendor address, and FSSAI compliance license number.

System 2 NEW v1.1

rPET Plastic Recycle Declaration

QR code maps to `/rpet`. Consumers enter the full bottle neck code (e.g. `MBB306C`) to verify the recycled plastic content percentage, manufacturing location, and PWM registration number — as mandated by Plastic Waste Management Rules.

Metric	Target	Metric	Target
Consumer page load	< 3 seconds on 4G	API response time	< 500ms P95
End-to-end QR flow	< 10 seconds	Admin onboarding	< 30 minutes

2. Problem Statement

Consumer Problem — FSSAI

Consumers have no easy way to verify rPET product authenticity, manufacturer details, or FSSAI compliance. Regulatory requirements mandate this information be accessible via QR code.

Consumer Problem — PWM Rules

India's Plastic Waste Management Rules require manufacturers to declare the recycled plastic content (rPET %) on packaging and make it verifiable. A QR-accessible declaration fulfils this mandate.

Regulatory Team Problem: The existing vendor management system (RLL/RCPM) cannot be modified. A new independent system is needed that the regulatory team can update without IT dependency for each change.

Solution: Dual-system web application within a single monorepo. System 1 handles FSSAI batch code lookups. System 2 handles PWM recycle content declarations. Both share a single admin portal with separate sections. Each maps to its own QR Planet QR code URL.

3. Stakeholders

Stakeholder	Role	Primary Need

Consumers	End users scanning QR code on bottle	Fast, mobile-friendly product info (both systems)
Nidhi (Regulatory Team)	Data manager — adds/edits all records	Simple admin forms, no code changes required
RCPL IT (Ashcharna/Pankaj)	Technical reference for existing system	Zero changes to RLL/RCPM systems
Ashish Chandorkar	Developer & system owner	Clear requirements, two URLs for QR Planet mapping
QR Planet	QR code provider	Two landing page URLs — one per QR system
PWM Compliance Authority	Regulatory body for plastic waste rules	Verifiable recycled content declaration on packaging

4. User Stories

System 1 — FSSAI Consumer Stories

ID	As a…	I want to…	So that…
US-01	Consumer	Scan QR and see manufacturer details	I can verify product authenticity
US-02	Consumer	Enter only 1-2 chars from bottle neck	The process is quick on mobile
US-03	Consumer	See FSSAI license, vendor name, address	I have complete compliance information
US-04	Consumer	Page loads fast on mobile data	I don't wait more than 3 seconds

System 2 — rPET Declaration Consumer Stories NEW v1.1

ID	As a…	I want to…	So that…
US-09	Consumer	Scan QR and enter the full bottle neck code	I can verify the recycled plastic content of my bottle
US-10	Consumer	See the rPET recycled content percentage clearly	I can make an informed environmental choice
US-11	Consumer	See the manufacturing location and PWM registration	I have full compliance details for the plastic used

Regulatory Team Stories (Both Systems)

ID	As a…	I want to…	So that…
US-05	Regulatory team member	Add new rPET product records (System 1)	Consumer FSSAI info stays current
US-06	Regulatory team member	Edit FSSAI license or address	Records are always accurate
US-07	Regulatory team member	Deactivate discontinued products	Outdated products don't show to consumers
US-08	Regulatory team member	Search products by vendor/category	I can find records quickly
US-12	Regulatory team member	Add/edit rPET recycle declarations (System 2)	PWM declaration data is always up to date
US-13	Regulatory team member	Set the recycle content % for each bottle SKU	Consumers see the correct rPET % for their product
US-14	Regulatory team member	Manage PWM registration numbers per declaration	Compliance records are maintained in one place

5. Functional Requirements

System 1 — FSSAI Product Information

P0 — Must Have (Launched)

ID	Requirement
FR-01	Consumer landing page at `/` with RCPL branding, instructions, and batch code input
FR-02	Batch code entry: 1–10 alphanumeric characters, case-insensitive, auto-uppercase transform
FR-03	Product info display: Product Name, Category, Subcategory, Vendor Name, Address, Manufacturer, FSSAI License, SKU, Status
FR-04	Admin form to CREATE product records with all required fields and FSSAI 14-digit validation
FR-05	Admin form to EDIT records (`batch_code` immutable after creation)
FR-06	Inactive product codes return friendly "not available" message

FR-07	Non-existent codes return "product not found" with clear guidance and back button
FR-08	System provides standalone URL at `/` for QR Planet mapping
FR-09	Admin product list with search (vendor name, category, batch code) and pagination (15 per page)
FR-10	Admin soft-delete (deactivate) with confirmation modal
FR-11	FSSAI validation: exactly 14 digits, numeric only
FR-12	Rate limiting on public endpoint (100 req / 15 min per IP)
FR-13	Health check endpoint at `/health`

System 2 — rPET Plastic Recycle Declaration NEW v1.1

P0 — Must Have

ID	Requirement
FR-20	Consumer landing page at `/rpet` with green recycling theme, instructions for finding bottle neck code, and code input field
FR-21	Bottle neck code format: `[VendorCode][ProductVariantCode][SKUCode][RecycleCode]` — e.g. `MBB306C`, alphanumeric, auto-uppercase, up to 20 characters
FR-22	Declaration info page at `/rpet/product/:code` displaying: Product Name, SKU, Manufacturing Location (full address), Recycle Content (% with progress bar), PWM Registration Number
FR-23	Visual recycled content indicator — animated progress bar showing rPET percentage
FR-24	Declaration lookup returns only `active` records; inactive returns friendly error
FR-25	Admin CRUD for declarations: create, edit, deactivate, paginated list, search
FR-26	`combined_code` is UNIQUE and immutable after creation
FR-27	Admin validates: `recycle_content` is numeric 0–100, all required fields present

FR-28	System provides standalone URL at `/rpet` for a second QR Planet mapping

P2 Future Sprints (Both Systems)

ID	Requirement
FR-15	Admin portal authentication (login credentials for regulatory team)
FR-16	Audit log of all admin changes (who changed what, when)
FR-17	Bulk import via CSV for initial data migration
FR-18	QR code generation preview in admin portal
FR-19	Lookup code tables (Vendor / Product Variant / SKU / Recycle codes) manageable from admin UI

6. Non-Functional Requirements

Category	Requirement
Performance	Consumer pages: < 3s on 4G mobile. API lookup: < 500ms P95 for both systems
Mobile-first	All consumer-facing pages optimised for 375px+ (iPhone SE) and upward
Security	Helmet.js headers, input sanitisation (express-validator), parameterised queries, rate limiting on all public endpoints
Availability	99% uptime target; JSON file DB with synchronous read/write, in-memory cache
Independence	Zero dependencies on existing RLL/RCPM vendor management system
URL Stability	Both `/` and `/rpet` URLs must remain stable once mapped to QR Planet
Data Integrity	`batch_code` and `combined_code` are UNIQUE (case-insensitive); validated at both API and form layers
Portability	No native Node.js addons — pure JavaScript dependencies only (solves Node 24 / Xcode build issues)

7. System Architecture

```
┌──────────────────────────────────────────────────────────────────┐
│  QR Planet (3rd party)                                           │
│  QR Code A ──maps to──▶ https://[domain]/        (System 1 — FSSAI) │
│  QR Code B ──maps to──▶ https://[domain]/rpet    (System 2 — PWM)   │
```

```
                                  │ Consumer scans QR code
                                  ▼
┌──────────────────────────────────────────────────────────────────────┐
│  FRONTEND  (React 18 + TypeScript + Tailwind CSS, Vite, port 5173)  │
│                                                                      │
│  SYSTEM 1 — FSSAI                                                    │
│  /                  ──▶ Landing Page (batch code search, blue theme) │
│  /product/:code     ──▶ Product Info Page (FSSAI compliance details) │
│                                                                      │
│  SYSTEM 2 — rPET DECLARATION (NEW)                                   │
│  /rpet              ──▶ Landing Page (bottle code search, green)     │
│  /rpet/product/:code──▶ Declaration Info Page (recycled content %)  │
│                                                                      │
│  SHARED ADMIN PORTAL                                                 │
│  /admin/products/*  ──▶ FSSAI product CRUD                          │
│  /admin/rpet/*      ──▶ rPET declaration CRUD                       │
└──────────────────────────────────────────────────────────────────────┘
                                  │ REST API over JSON
                                  ▼
┌──────────────────────────────────────────────────────────────────────┐
│  BACKEND  (Node.js + Express + Helmet + Rate Limiter, port 3001)    │
│                                                                      │
│  SYSTEM 1 PUBLIC                                                     │
│  GET  /api/products/lookup?code=AA        ◀── Consumer lookup       │
│                                                                      │
│  SYSTEM 2 PUBLIC (NEW)                                               │
│  GET  /api/rpet/lookup?code=MBB306C       ◀── Consumer lookup       │
│                                                                      │
│  ADMIN                                                               │
│  GET/POST/PUT/DELETE /api/admin/products  ◀── FSSAI CRUD            │
│  GET/POST/PUT/DELETE /api/admin/rpet      ◀── rPET declaration CRUD │
│                                                                      │
│  GET  /health                             ◀── Monitoring            │
└──────────────────────────────────────────────────────────────────────┘
                                  │ fs.readFileSync / writeFileSync
                                  ▼
┌──────────────────────────────────────────────────────────────────────┐
│  DATABASE  (JSON file — rpet.json, in-memory cache)                 │
│  Collection: products      (FSSAI product records)                  │
│  Collection: declarations  (rPET recycle declaration records)        │
│  No native modules — pure JavaScript, runs on any Node.js version   │
└──────────────────────────────────────────────────────────────────────┘
```

Tech Stack: React 18 · TypeScript (strict) · Tailwind CSS · Vite · @tanstack/react-query · React Router v6 · Node.js · Express.js · express-validator · Helmet.js · express-rate-limit · GitHub Actions CI/CD

8. Data Model

Collection: `products` (System 1 — FSSAI)

Field	Type	Required	Description
id	number	Auto	Internal auto-increment identifier
batch_code	string	Yes	UNIQUE (case-insensitive). Printed on bottle neck — e.g. "AA". Max 10 chars, alphanumeric.
category	string	Yes	Product category — e.g. "Dairy"
subcategory	string	Yes	Product subcategory — e.g. "Full Cream Milk"
vendor_name	string	Yes	Manufacturing unit / vendor name
vendor_address	string	Yes	Full address with PIN code
fssai_license	string	Yes	Exactly 14-digit FSSAI license number
sku	string	No	Stock Keeping Unit code
product_name	string	No	Full product name
manufacturer	string	No	Parent company / manufacturer name
status	'active' | 'inactive'	Auto	Default: 'active'. Soft-delete sets to 'inactive'.
created_at	string (ISO 8601)	Auto	Record creation timestamp
updated_at	string (ISO 8601)	Auto	Last modification timestamp

Collection: `declarations` (System 2 — rPET Declaration) NEW v1.1

Field	Type	Required	Description
id	number	Auto	Internal auto-increment identifier
combined_code	string	Yes	UNIQUE (case-insensitive). Bottle neck code combining VendorCode + ProductVariantCode + SKUCode + RecycleCode. E.g. "MBB306C". Max 20 chars, alphanumeric.
product_name	string	Yes	Full product name — e.g. "RCPL rPET Water Bottle 500ml"
sku	string	Yes	Stock Keeping Unit — e.g. "RCPL-PET-500"
manufacturing_location	string	Yes	Full manufacturing address including city, state, and PIN code

recycle_content	number (0–100)	Yes	Recycled plastic content as percentage — e.g. 100 for 100% rPET
pwm_reg_no	string	Yes	PWM (Plastic Waste Management) Registration Number — e.g. "PWM/MH/2024/001234"
status	'active' | 'inactive'	Auto	Default: 'active'. Soft-delete sets to 'inactive'.
created_at	string (ISO 8601)	Auto	Record creation timestamp
updated_at	string (ISO 8601)	Auto	Last modification timestamp

Bottle Code Structure (System 2):
The `combined_code` is a concatenation of four lookup segments:

```
[ VendorCode ][ ProductVariantCode ][ SKUCode ][ RecycleCode ]
```

Example: **`MBB306C`** = Vendor `M` + Variant `BB` + SKU code `306` + Recycle code `C`
Each segment maps to a lookup table administered by the regulatory team.

DB File Schema (rpet.json)

```
{
  "products": [ ...RpetProduct[] ],
  "nextId": 4,
  "declarations": [ ...RpetDeclaration[] ],
  "declNextId": 3
}
```

9. API Specification

System 1 — FSSAI Public Endpoint

Method	Path	Auth	Description
GET	/api/products/lookup?code={batchCode}	None	Consumer product lookup by batch code
GET	/health	None	Service health check

Response	Status	Condition
`{ success: true, data: RpetProduct }`	200	Valid active code found
`{ success: false, error: "No product found..." }`	404	Code does not exist or is inactive
`{ success: false, error: "validation message" }`	400	Invalid code format

System 2 — rPET Declaration Public Endpoint NEW v1.1

Method	Path	Auth	Description
GET	/api/rpet/lookup?code={combinedCode}	None	Consumer declaration lookup by bottle neck code

Response	Status	Condition
`{ success: true, data: RpetDeclaration }`	200	Valid active code found
`{ success: false, error: "No declaration found..." }`	404	Code does not exist or is inactive
`{ success: false, error: "validation message" }`	400	Invalid code format

Admin Endpoints — System 1 (FSSAI Products)

Method	Path	Description
GET	/api/admin/products?page=1&limit=20	List all products (paginated, sorted by created_at desc)
GET	/api/admin/products/search?q=query	Search products by vendor name, category, or batch code
GET	/api/admin/products/:id	Get single product by ID
POST	/api/admin/products	Create product — returns 201 with new record
PUT	/api/admin/products/:id	Update product (batch_code field ignored in updates)
DELETE	/api/admin/products/:id	Soft-delete — sets status to 'inactive'

Admin Endpoints — System 2 (rPET Declarations) NEW v1.1

Method	Path	Description
GET	/api/admin/rpet?page=1&limit=20	List all declarations (paginated, sorted by created_at desc)
GET	/api/admin/rpet/search?q=query	Search by combined_code, product_name, or SKU
GET	/api/admin/rpet/:id	Get single declaration by ID

POST	/api/admin/rpet	Create declaration — returns 201 with new record
PUT	/api/admin/rpet/:id	Update declaration (combined_code field ignored in updates)
DELETE	/api/admin/rpet/:id	Soft-delete — sets status to 'inactive'

Shared API Conventions

Convention	Detail
Response envelope	`{ success: boolean, data?: T, error?: string, message?: string }`
Rate limiting	Public: 100 req/15 min. Admin: 300 req/15 min (per IP)
Validation errors	400 with first failing validation message
Not found	404 via NotFoundError middleware
Conflict (duplicate code)	409 with descriptive message
Code normalisation	All lookup codes converted to UPPERCASE before DB comparison

10. UI/UX Requirements

Page	URL	Key Elements	Design Notes
FSSAI Landing Page	`/`	RCPL logo, 3-step instruction box, batch code input, "Get Product Details" button	Mobile-first, blue gradient bg (#0D47A1→#1976D2), white card, monospace input, auto-uppercase
FSSAI Product Info	`/product/:code`	Back button, batch code header, product name hero, 3 info cards, status badge, loading skeleton	Cards: Product Details · Manufacturer Details · Compliance. Shimmer skeleton.
rPET Landing Page	`/rpet`	Recycle icon (♻), instructions with code structure diagram, bottle code input, "View Recycle Declaration" button	Green theme (green-900 → teal-700 gradient). Shows code breakdown: Vendor + Variant + SKU + Recycle segments.
rPET Declaration Info	`/rpet/product/:code`	Back button, product name hero, recycle % badge, animated progress bar, manufacturing card, PWM compliance card	Green gradient header. Hero card shows ♻ icon + large % number. Progress bar animates on load.

Admin Portal	`/admin/*`	Collapsible sidebar with two nav sections (FSSAI Products / rPET Declarations), product/declaration tables, create/edit forms, confirmation modal, toasts, footer links to both consumer pages	Desktop-optimised. Blue (FSSAI section) and green (rPET section) active states in sidebar. Inline form validation.

11. Success Metrics

Metric	Target	Measurement Method
Consumer page load time (both systems)	< 3 seconds on 4G	Lighthouse / browser DevTools
API lookup response P95 (both systems)	< 500ms	Server logs / monitoring
QR scan → info display	< 10 seconds	Manual end-to-end test
Admin onboarding time	< 30 minutes	User observation with Nidhi
Admin form error rate	< 2% of submissions	Validation error logs
Declaration records managed (System 2)	All active SKUs entered by Nidhi before QR mapping	Admin portal record count

12. Out of Scope

- Integration with existing RLL/RCPM vendor management system
- Consumer authentication or account creation
- QR code generation (handled by QR Planet)
- Email/SMS notifications
- Multi-language support (Hindi, regional languages)
- Native mobile app
- Price or inventory information
- Dynamic recycle code lookup table management via UI (P2 item — currently hardcoded seed data)

13. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation

QR Planet URL mapping delay	Low	High	Provide both URLs early; test with direct URL first
Regulatory team needs more fields	Medium	Medium	JSON schema is flexible — add fields without migration scripts
Mobile network failures	Medium	High	Error states with retry; loading skeletons for perceived performance
JSON file DB concurrency under high load	Low	Medium	In-memory cache + synchronous writes; upgrade to PostgreSQL if traffic grows beyond prototype scale
Admin portal misuse (no auth)	Medium	Medium	P2: Add login auth in Sprint 2 before public launch
Combined code collision (System 2)	Low	Low	UNIQUE constraint on combined_code; 409 conflict response on duplicate
PWM regulation field changes	Low	Medium	Declaration schema uses nullable optional fields; easy to extend without breaking existing records

14. Acceptance Criteria

System 1 — FSSAI

- ☐ Consumer can scan QR, enter "AA", see full FSSAI product details within 10 seconds on mobile
- ☐ Inactive product code returns "not available" (not a raw error)
- ☐ Non-existent code returns friendly "product not found" with back button
- ☐ Admin can create a new product record with all required fields
- ☐ Admin can edit vendor address and FSSAI license number
- ☐ Admin can deactivate a product (soft delete with confirmation modal)
- ☐ FSSAI validation rejects anything that is not exactly 14 digits

System 2 — rPET Declaration NEW v1.1

- ☐ Consumer can scan QR, enter "MBB306C", see recycle declaration with % and PWM details within 10 seconds
- ☐ Recycle content progress bar renders correctly for values between 0% and 100%
- ☐ Inactive declaration code returns friendly error with "Try Again" button
- ☐ Admin can create a new declaration with combined_code, product name, manufacturing location, recycle %, and PWM reg no
- ☐ Admin can edit recycle content and PWM registration number
- ☐ Admin can deactivate a declaration (soft delete with confirmation modal)
- ☐ Duplicate combined_code on creation returns 409 conflict error in admin form
- ☐ Admin sidebar shows both "FSSAI Products" and "rPET Declarations" sections

- ☐ Admin footer provides quick links to both consumer-facing pages

Shared

- ☐ All pages render correctly on iPhone SE (375px) screen width
- ☐ CI/CD pipeline passes: lint → test → build → security audit
- ☐ Both landing page URLs are provided for QR Planet mapping
- ☐ GitHub repository is up to date with all changes pushed to main

15. Delivery Plan

Sprint 1 — System 1: FSSAI (Completed)

0. **Foundation** — CLAUDE.md, package.json, tsconfig, .gitignore, CI/CD pipeline
1. **Backend API** — Express + JSON file DB + all FSSAI endpoints (lookup + admin CRUD)
2. **Consumer Frontend** — Landing Page + Product Info Page (mobile-first, React + Tailwind, blue theme)
3. **Admin Portal** — Product list, create/edit forms, pagination, search, toasts
4. **Testing** — Unit + integration tests for frontend (Vitest + @testing-library) and backend
5. **GitHub + CI/CD** — Repository created, initial commit pushed, Actions pipeline running

Sprint 2 — System 2: rPET Declaration (Completed in v1.1)

1. **Backend Extension** — DB schema extended with `declarations` collection. New routes: `/api/rpet/lookup` and `/api/admin/rpet` (full CRUD)
2. **Consumer Pages** — `/rpet` landing page (green theme, code structure guide) + `/rpet/product/:code` declaration info page with animated recycle progress bar
3. **Admin Extension** — `/admin/rpet` list page + `/admin/rpet/new` & edit forms. Admin sidebar updated with two nav sections. Footer links to both consumer pages.
4. **Seeded Demo Data** — 2 sample rPET declarations: MBB306C (100% rPET, 500ml) and MBB312A (75% rPET, 1L juice)
5. **PRD Updated & Committed** — PRD v1.1 documenting both systems. All changes pushed to GitHub.

Sprint 3 — Hardening (Planned)

6. **Admin Authentication** — Login screen + session management for regulatory team (P2)
7. **Production Deployment** — URL delivery to QR Planet for both QR codes; uptime monitoring

8 **Regulatory Team Walkthrough** — Demo with Nidhi; onboarding for both admin sections